INFOSYS SPRINGBOARD ONLINE INTERNSHIP

HotelRevAI

AI-Driven Revenue Analysis for Hotels

Final Project Report – Milestone 4

Submitted By:
A Veda Varshini
Computer Science and Engineering (Data Science)
Vellore Institute of Technology (VIT), Vellore

To:

Mentor: Selciya
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1. Abstract

The hospitality industry generates large volumes of booking data that contain valuable insights related to customer behavior, revenue trends, and demand patterns. However, raw booking data often contains inconsistencies, missing values, and redundant information, making it difficult to analyze and extract meaningful insights.

The objective of this project is to analyze hotel booking data and generate revenue insights using **Power BI and data analytics techniques**. The dataset used in this project was obtained from Kaggle and includes booking details such as customer type, stay duration, arrival date, country, cancellation status, and average daily rate (ADR).

The project involves **data cleaning, transformation, data modeling, DAX calculations, and dashboard development**. Multiple business metrics were created to evaluate revenue performance, booking trends, customer loyalty, and occupancy behavior.

In the final stage of the project (Milestone-4), the dashboards were further enhanced by improving **visual design, color consistency, professional layout, and analytical clarity**. Additionally, a new analytical dashboard titled **Seasonality & Stay Pattern Analysis** was introduced to provide deeper insights into stay duration patterns and seasonal demand behavior.

Overall, the system converts raw booking data into **interactive visual dashboards that support data-driven decision making in hotel revenue management**.

2. Introduction

Revenue management plays a critical role in the hospitality industry. Hotels rely on accurate demand forecasting, customer behavior analysis, and pricing strategies to maximize profitability and maintain optimal occupancy levels.

However, hotel booking data is often stored in large datasets containing thousands of records. These datasets typically include information such as booking dates, customer types, room types, cancellations, and guest counts. Without proper data analysis tools, it becomes difficult to identify patterns and trends that can support strategic decision-making.

This project aims to transform raw hotel booking data into meaningful insights using **data analytics and visualization techniques**. Power BI is used as the primary analytics platform to clean the dataset, create calculated metrics, and develop interactive dashboards.

The project focuses on analyzing key aspects of hotel operations such as:

- Revenue generation patterns
- Customer booking behavior
- Seasonal demand trends
- Booking lead time patterns
- Customer loyalty and repeat bookings

By converting raw data into visual dashboards, hotel management can gain a deeper understanding of their business performance and make informed decisions to improve revenue and operational efficiency.

3. Problem Statement

Hotels generate large volumes of booking data that contain valuable insights about customer behavior, revenue trends, and demand patterns. However, raw booking datasets often contain missing values, redundant information, and inconsistent formats.

Without proper data preparation and visualization tools, it becomes difficult for hotel management to analyze:

- Revenue fluctuations
- Customer booking behavior
- Seasonal demand patterns
- Impact of cancellations
- Customer retention trends

This project addresses these challenges by developing a **Power BI based analytical system** that cleans, models, and visualizes hotel booking data to generate actionable insights for revenue optimization.

4. Objectives

The primary objectives of this project are:

- To analyze hotel booking data and understand revenue trends.
- To identify key customer segments contributing to hotel revenue.
- To analyze booking patterns based on lead time and stay duration.
- To evaluate the impact of cancellations on revenue.
- To measure customer loyalty through repeat booking analysis.
- To develop interactive dashboards using Power BI for business insights.
- To support data-driven decision making in hotel revenue management.

5. Dataset Description

The dataset used in this project was obtained from **Kaggle – Hotel Booking Dataset**. The dataset contains booking information for two types of hotels:

- City Hotel
- Resort Hotel

The dataset includes multiple attributes related to booking behavior and customer information.

Key dataset attributes include:

- Hotel type
- Arrival date
- Lead time
- Customer type
- Country
- Market segment
- Distribution channel
- Average Daily Rate (ADR)
- Number of adults, children, and babies
- Number of weekend nights
- Number of weekday nights
- Booking cancellation status

This dataset provides valuable information for analyzing **customer behavior, booking patterns, and revenue generation**.

6. Data Loading and Pre-Processing

The dataset was imported into **Power BI using a CSV file**. Data preparation was performed using **Power Query Editor**, which allowed the dataset to be cleaned and transformed before loading it into the data model.

The following preprocessing operations were performed:

- Handling missing or null values
- Removing duplicate records
- Correcting data types (date, numeric, text)
- Removing irrelevant columns
- Standardizing column names

These preprocessing steps ensured that the dataset was accurate and suitable for further analysis.

7. Feature Engineering

Several new analytical columns were created to support deeper analysis.

New calculated columns include:

Total Nights

Total Nights = Weekend Nights + Weekday Nights

Total Guests

Total Guests = Adults + Children + Babies

Revenue

Revenue = ADR × Total Nights

Arrival Date

Multiple arrival date columns were merged into a single date column for time-based analysis.

These derived features helped create meaningful metrics for revenue and occupancy analysis.

8. Data Modelling

Data modeling was implemented in Power BI to establish relationships between tables and support efficient data analysis.

The model includes:

Fact Table

Hotel_Data – Contains booking transaction records.

Dimension Table

Calendar Table – Contains:

- Date
- Month
- Month Number
- Year

A **one-to-many relationship** was created:

Calendar[Date] → Hotel_Data[Arrival Date]

This relationship allows accurate time-based filtering across dashboards.

9. DAX Measures

Several DAX measures were created to calculate business metrics.

Examples include:

Total Revenue

Total Revenue = SUM(Hotel_Data[Revenue])

Total Bookings

COUNTROWS(Hotel_Data)

Total Guests Count

SUM(Hotel_Data[Total Guests])

Revenue per Night

Total Revenue ÷ Total Nights Stayed

Cancellation Rate

Cancelled Bookings ÷ Total Bookings

Repeat Guest Percentage

Bookings Repeated ÷ Total Bookings

These DAX measures help convert raw data into meaningful analytical metrics.

10. Business Metrics

The following key business metrics were developed.

Revenue Metrics

- Total Revenue
- Revenue per Night
- Revenue Trend
- Revenue Loss due to cancellation

Booking Metrics

- Total Bookings
- Lead Time Analysis
- Booking Demand Patterns

Customer Metrics

- Total Customers
- Repeated Customers
- Repeat Guest Percentage

Occupancy Metrics

- Occupancy Proxy
- Total Nights Stayed

Market Metrics

- Revenue by Market Segment
- Revenue by Distribution Channel
- Bookings by Country

These metrics provide insights into **hotel performance, customer behavior, and demand trends**.

11. Dashboard Implementation

Four interactive dashboards were developed using Power BI.

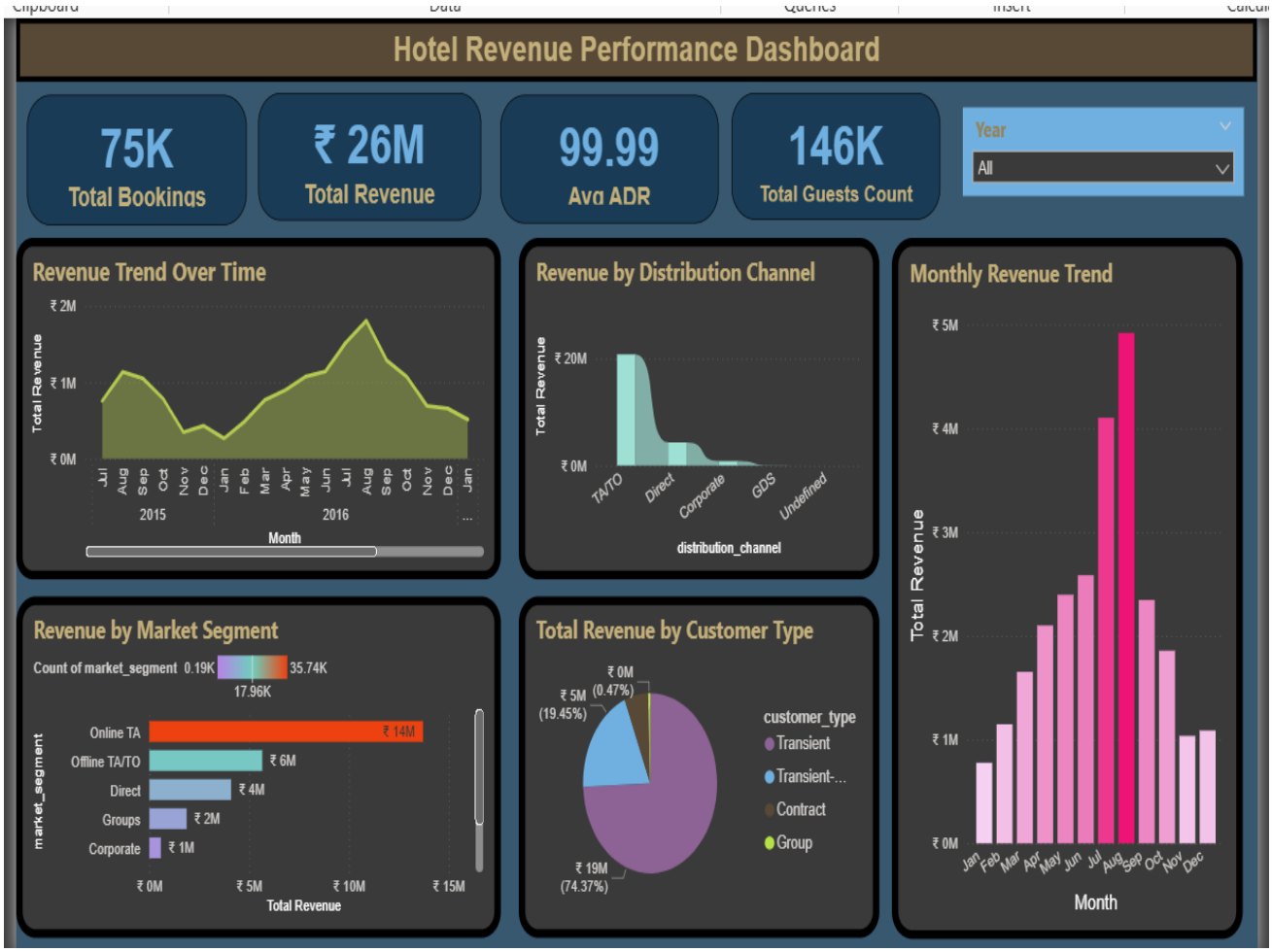
Dashboard 1: Hotel Revenue Performance

This dashboard provides an overview of hotel revenue performance using key KPIs:

- Total Bookings
- Total Revenue
- Average Daily Rate (ADR)
- Total Guest Count

Visualizations include:

- Revenue Trend Over Time
- Revenue by Distribution Channel
- Monthly Revenue Trend
- Revenue by Market Segment
- Revenue by Customer Type



Key insight:
Online travel agencies contribute significantly to revenue generation.

Dashboard 2: Customer and Booking Behaviour

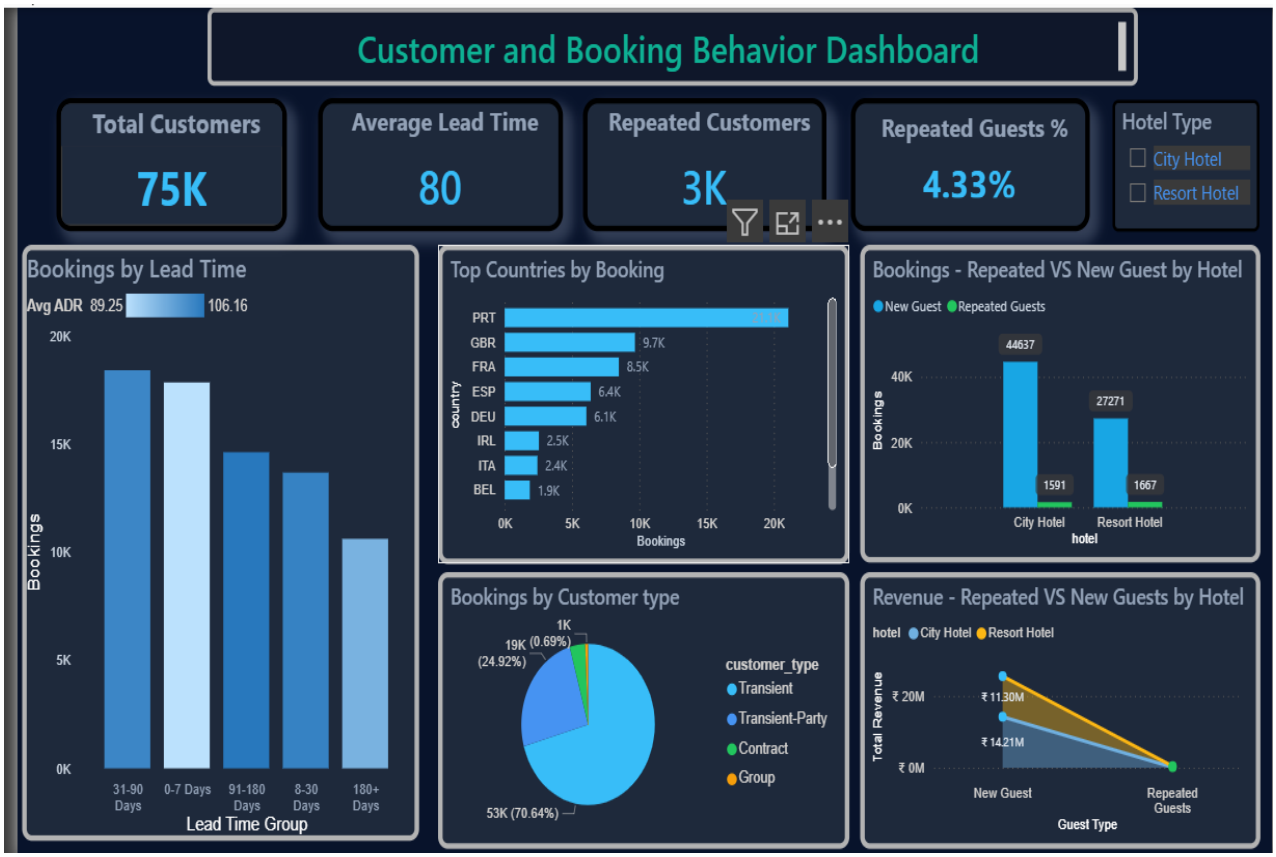
This dashboard focuses on understanding customer booking behavior.

KPIs:

- Total Customers
- Average Lead Time
- Repeated Customers
- Repeated Guest %

Visualizations:

- Bookings by Lead Time
- Top Countries by Booking
- New vs Repeated Guests
- Bookings by Customer Type



Key insight:
Customer loyalty is relatively low, with only a small percentage of repeat guests.

Dashboard 3: Revenue Optimization and Key Insights

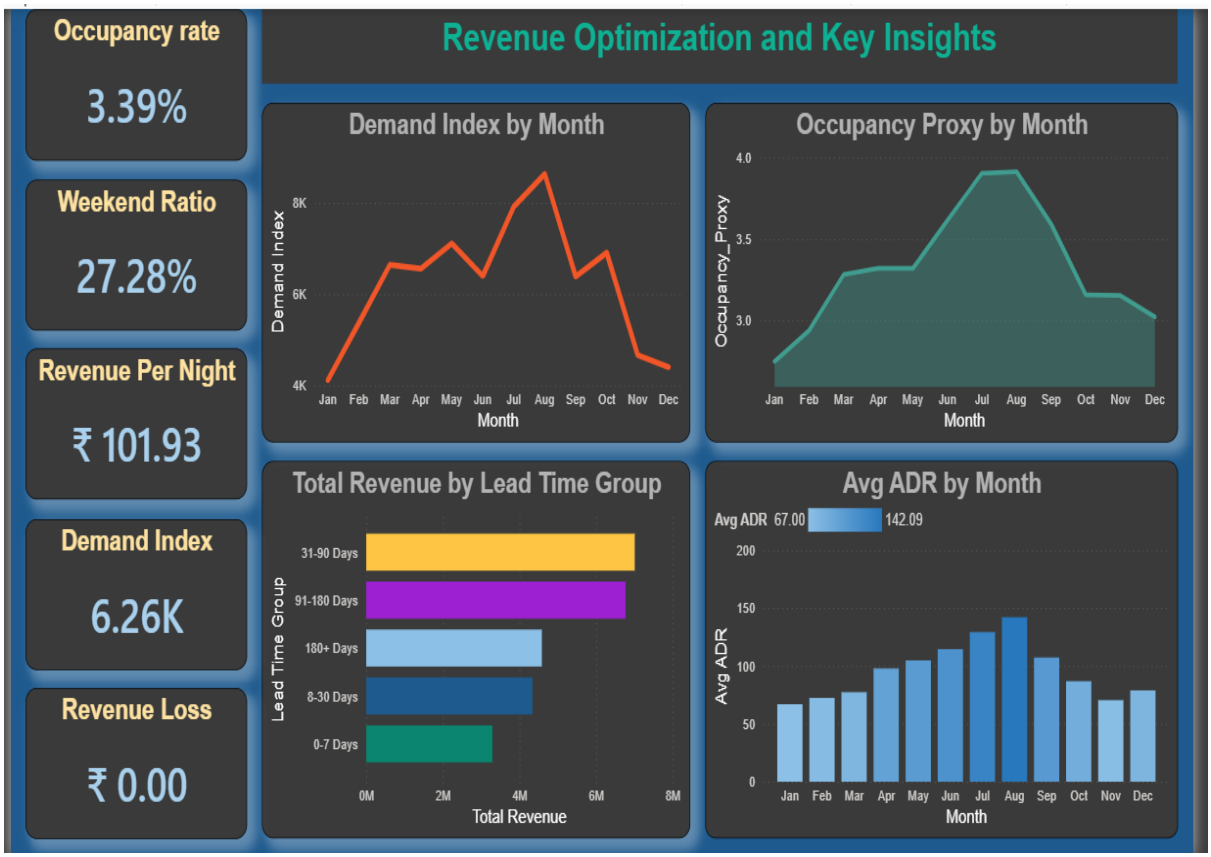
This dashboard focuses on revenue performance and operational efficiency.

KPIs include:

- Occupancy Rate
- Weekend Ratio
- Revenue per Night
- Demand Index
- Revenue Loss

Visualizations:

- Demand Index by Month
- Occupancy Proxy by Month
- Revenue by Lead Time
- ADR Trend by Month



These insights help hotels optimize pricing and demand management.

Dashboard 4: Seasonality & Stay Pattern Analysis (Added in Milestone-4)

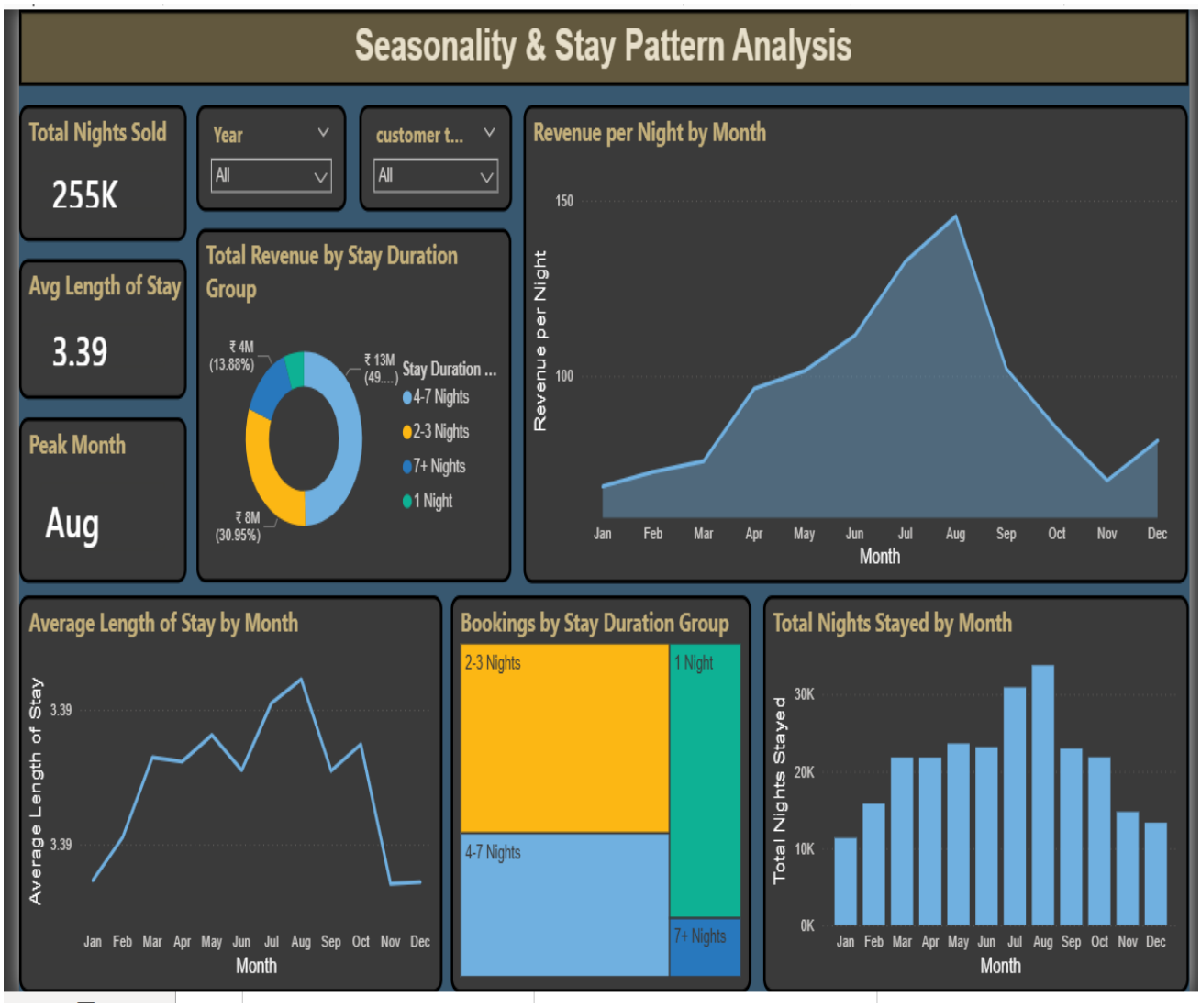
This dashboard was introduced in the final milestone to analyze seasonal demand and stay patterns.

KPIs include:

- Total Nights Sold
- Average Length of Stay
- Peak Month

Visualizations include:

- Revenue per Night by Month
- Average Length of Stay Trend
- Total Nights Stayed by Month
- Bookings by Stay Duration Group



Key insight:
August appears as the peak demand month with higher revenue and longer stay durations.

12. Key Insights

The analysis reveals several important insights:

- Online travel agencies generate the highest booking revenue.
- Transient customers contribute the majority of bookings.
- Repeat guest percentage is relatively low.
- August shows the highest seasonal demand.
- Early bookings (31–90 days lead time) contribute significantly to revenue.

These insights can help hotels improve pricing strategies and marketing campaigns.

13. Improvements in Milestone-4

In the final milestone, several improvements were made to enhance the dashboards:

- Improved professional color themes
- Better layout alignment and visual clarity
- Enhanced KPI card design
- Improved readability of charts
- Addition of a new dashboard for seasonal demand analysis

These improvements make the dashboards more professional and easier to interpret.

14. Conclusion

The HotelRevAI project successfully demonstrates how data analytics and visualization can transform raw hotel booking data into meaningful business insights.

By implementing structured data modeling, DAX calculations, and interactive dashboards, the system enables detailed analysis of revenue performance, customer behavior, and seasonal demand patterns.

The final dashboards provide valuable insights that can help hotel management:

- Improve revenue optimization strategies
- Understand customer behavior patterns
- Forecast seasonal demand
- Enhance operational decision making

Overall, the project highlights the importance of **data-driven decision making in the hospitality industry**.

15. Future Enhancement

Future improvements to the system may include:

• Machine learning models for demand forecasting.
• Real-time live booking pipeline.
• Customer segmentation modelling using clustering.
• Competitor rate benchmarking.
• Automated anomaly detection alerts.

These improvements could further enhance hotel revenue management capabilities.

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